

RESEARCH ARTICLE | MAY 14 2026

Stockmayer fluid with a shifted dipole. I. Bulk behavior

Pierre J. Walker ; Ananya Venkatachalam ; Samuel Varner ; Bilin Zhuang ; Zhen-Gang Wang  



J. Chem. Phys. 164, 184512 (2026)

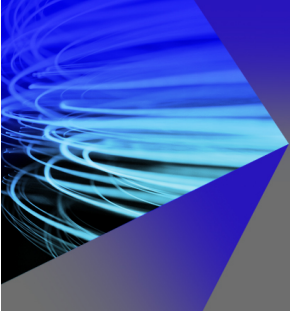
<https://doi.org/10.1063/5.0331053>



Articles You May Be Interested In

Stockmayer fluid with a shifted dipole. II. Interfacial behavior

J. Chem. Phys. (May 2026)



AIP Advances

Why Publish With Us?



21DAYS
average time
to 1st decision



OVER 4 MILLION
views in the last year



INCLUSIVE
scope

[Learn More](#)



Stockmayer fluid with a shifted dipole. I. Bulk behavior

Cite as: J. Chem. Phys. 164, 184512 (2026); doi: 10.1063/5.0331053

Submitted: 23 February 2026 • Accepted: 23 April 2026 •

Published Online: 14 May 2026



Pierre J. Walker,¹ Ananya Venkatachalam,² Samuel Varner,¹ Bilin Zhuang,^{2,a)}
and Zhen-Gang Wang^{1,b)}

AFFILIATIONS

¹ Division of Chemistry and Chemical Engineering, California Institute of Technology, Pasadena, California 91125, USA

² Department of Chemistry, Harvey Mudd College, Claremont, California 91711, USA

^{a)} Electronic mail: bzhuang@g.hmc.edu

^{b)} Author to whom correspondence should be addressed: zgwang@caltech.edu

ABSTRACT

Shifting the point dipole from the center of a Stockmayer particle is a simple geometric modification, whose implications on liquid structure, dielectric response, and phase behavior remain incompletely understood. We combine molecular dynamics simulations with analytical theory to provide a unified physical interpretation of how dipole shifts reshape microscopic correlations and propagate to macroscopic properties. We show that dipole-shifting breaks the fore-aft symmetry of the local electrostatic field, producing only modest changes in radial packing but strong alterations in angular structure within the first solvation shell. Enhanced alignment near the dipole head is accompanied by frustrated orientational correlations near the tail, leading to broader angular distributions and a shift away from axial configurations at strong coupling. These structural asymmetries weaken cooperative ordering and result in a systematic reduction in the dielectric constant, despite locally stronger interactions. For large shifts, the dielectric response approaches the Debye limit, indicating effective suppression of dipole-dipole correlations. The same geometric frustration governs vapor-liquid equilibria: while increasing dipole strength raises the critical temperature, even modest shifts disrupt the highly polarized liquid state that emerges at strong coupling and can suppress ferroelectric-like ordering. Predictions from a reparameterized co-oriented fluid functional equation for electrostatic interactions theory capture these trends within its domain of validity, highlighting the connection between local structure and macroscopic observables. Overall, this work demonstrates that dipole location, not only magnitude, provides a powerful control parameter in dipolar fluids and offers a clear framework for understanding geometric frustration in electrostatic liquids.

Published under an exclusive license by AIP Publishing. <https://doi.org/10.1063/5.0331053>

I. INTRODUCTION

Electrostatic interactions govern liquid structure and thermodynamic behavior across scales, from local solvation structures to macroscopic response functions, for example, the dielectric constant.¹ The cross-scale connection is vital for the interpretation and modeling of polar liquids and mixtures. This coupling between scales is especially important when one considers both the bulk behavior of solvents, such as the preferential solvation of ions,^{2,3} and behavior at liquid interfaces, where orientational ordering can generate local electric fields that bias reaction pathways.^{4–13} The present study focuses on the equilibrium bulk structure and dielectric response, where the connections between microscopic correlations and observables can be most clearly established.

The magnitude and physical origin of anomalously strong interfacial fields and their accompanying bulk properties remain actively debated, emphasizing the need for minimal, controllable models that enable rigorous study of electrostatics, packing, and orientational correlations.^{14,15} Dipolar model fluids provide a convenient system to isolate the effects of excluded-volume packing and orientational interactions. Among such models, the Stockmayer fluid is an especially effective minimal model of a polar liquid.^{16–18} Simulations and existing theory have used the Stockmayer model to connect microscopic pair correlations to collective behavior, including in regimes that exhibit strong orientational ordering at large dipole strengths.^{19–22} Stockmayer-type models have also been used to investigate liquid-vapor interfaces and the impact of external fields on surface polarization.^{23–26} Interactions in this model are

simple enough to interpret mechanistically and yet are able to capture nontrivial coupling between local structure and macroscopic observables.

A primary limitation of the Stockmayer fluid model is its treatment of molecules as perfect spheres with centrally located point dipoles. While mathematically convenient, this idealized symmetry fails to describe the inherent asymmetry of most polar molecules. In reality, few molecules exhibit such fore-aft symmetry; instead, they often possess off-center charge distributions, where the electrostatic and geometric centers do not coincide. To bridge this gap, the shifted-dipole Stockmayer fluid (sd-SF) model introduces a minimal source of electrostatic asymmetry by displacing the dipole away from the Lennard-Jones center, providing a more physically grounded representation of molecular geometry. This model was first introduced by Kusaka *et al.*²⁷ to examine the sign effect in ion-induced nucleation of water droplets in the atmosphere. Kantorovich and co-workers^{28–31} used a similar model fluid to examine cluster formation in ferrofluids, focusing on large shifts and high dipole strengths. Similar studies by Steinbach *et al.*^{32–36} have introduced anisotropies to represent magnetic colloidal particles, examining both equilibrium and non-equilibrium behavior. Particularly relevant to this study is the work by Langenbach and co-workers,^{37–42} who developed the Co-Oriented Fluid Functional Equation for Electrostatic Interactions (COFFEE) framework to model the fluid-phase behavior of the shifted-dipole Stockmayer fluid, using information from the orientational distribution functions to determine thermodynamic properties. For small dipolar shifts and strengths, this work has been used to study the dielectric properties^{39,40} and phase behavior,³⁸ even extending the work to mixtures,^{41,42} for this simple model fluid. Collectively, these studies establish that even moderate dipole shifts can substantially impact dielectric response and vapor–liquid equilibria of a dipolar fluid. Despite ongoing work, a fundamental gap remains in the physical intuition underlying the effect of introducing a dipole shift and the resulting macroscopic changes. Much of the existing literature emphasizes reproducing and predicting simulation results, while the underlying driving forces remain unresolved. For example, it is not clear whether increasing the magnitude or shift of the dipole increases or frustrates global order, or how local structural asymmetries translate into the dielectric response. In the present work, we focus on small dipole shifts and moderate dipole strengths, both because this regime is relevant to the range of asymmetries exhibited by common polar solvents and because it allows us to isolate how modest electrostatic asymmetry alters bulk structure.

We build on prior studies of sd-SF but shift the emphasis to developing an understanding of the bulk physics of this system. To this end, we leverage molecular dynamics simulations, as discussed in Sec. II B. We also evaluate the suitability of the COFFEE theory for studying the bulk behavior of sd-SF systems, as discussed in Sec. II C. We then examine the structural properties of the sd-SF across a wide range of dipole shifts and strengths, developing intuition for the changes in the orientational–radial distribution functions in Sec. III A. Using this intuition, we examine changes in the dielectric response and phase behavior of sd-SF fluids in Secs. III B and III C, respectively. We also note that this is a companion piece to a following study examining the vapor–liquid interfacial behavior in sd-SF systems.⁴³

II. METHODOLOGY

A. System

We consider a system of N particles at constant volume V and temperature T . An example particle i is schematically illustrated in Fig. 1, whose geometrical center is located at $\mathbf{r}_i$ and whose dipole moment $\boldsymbol{\mu}_i$ is shifted by a distance d from this center along the dipole vector. That is, the dipole of the particle is located at $\mathbf{r}_i^d = \mathbf{r}_i + \hat{\boldsymbol{\mu}}_i d$, where $\hat{\boldsymbol{\mu}}_i$ is a unit vector in the direction of $\boldsymbol{\mu}_i$. The particles interact through a truncated Lennard-Jones (LJ) interaction between the geometrical centers and a long-range dipole–dipole interaction between the displaced point dipoles. The LJ interaction is given by

$$\phi_{\text{LJ}}(r_{ij}) = \begin{cases} 4\epsilon \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^6 \right], & r_{ij} \leq r_c, \\ 0, & r_{ij} > r_c, \end{cases} \quad (1)$$

where ϵ is the pair interaction strength, σ is the characteristic particle diameter, and $r_{ij} = |\mathbf{r}_j - \mathbf{r}_i|$ is the distance between the geometrical centers of particles i and j . The LJ cutoff distance is set to $r_c = 2.5\sigma$, which is well beyond the potential minimum at $r_{\text{min}} = 2^{1/6}\sigma$.

The electrostatic interactions between the permanent point dipoles are described by the dipole–dipole potential,

$$\phi_{\text{dipole}}(\mathbf{r}_{ij}^d) = \frac{1}{4\pi\epsilon_0} \left[\frac{\boldsymbol{\mu}_i \cdot \boldsymbol{\mu}_j}{(r_{ij}^d)^3} - 3 \frac{(\boldsymbol{\mu}_i \cdot \mathbf{r}_{ij}^d)(\boldsymbol{\mu}_j \cdot \mathbf{r}_{ij}^d)}{(r_{ij}^d)^5} \right], \quad (2)$$

where ϵ_0 is the vacuum permittivity, $\mathbf{r}_{ij}^d = \mathbf{r}_j^d - \mathbf{r}_i^d$ is the vector connecting the displaced dipoles, and $r_{ij}^d = |\mathbf{r}_{ij}^d|$ is its magnitude. Here, $\boldsymbol{\mu}_i$ and $\boldsymbol{\mu}_j$ denote the dipole moments of particles i and j , respectively. This model is nearly equivalent to the original Stockmayer fluid model,¹⁶ with the exception of the dipole being shifted off the particle center. For the purposes of this work, the sd-SF model can

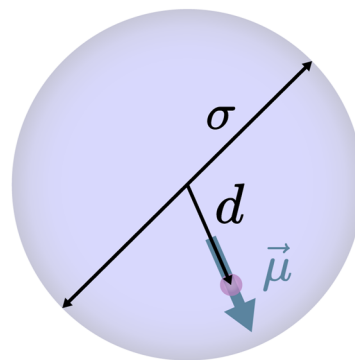


FIG. 1. Model of a single particle of the shifted-dipole Stockmayer fluid (sd-SF). Compared to the regular Stockmayer fluid, the particle has a dipole $\vec{\mu}$ shifted from the center by a distance d in the direction of the dipole (depicted as the blue arrow). The excluded-volume interaction between the particles is described by the Lennard-Jones potential with characteristic length scale σ , and the electrostatic interaction is the dipole–dipole interaction between the shifted dipoles.

be thought of as an explicit solvent where the background dielectric has been set to unity.

B. Molecular dynamics

1. Simulations

We perform molecular dynamics (MD) simulations of the shifted-dipole Stockmayer model in the NVT ensemble using the LAMMPS simulation package,⁴⁴ executed on graphics processing units (GPUs).^{45,46} Because no native shifted-dipole Stockmayer potential is available in LAMMPS, each molecule i is represented as a rigid dimer composed of two components (Fig. 1): an LJ particle with characteristic size σ and a massless, infinitesimally small “ghost” particle carrying a permanent dipole moment μ_i . The two sites are separated by a fixed bond length d , such that the dipole is displaced from the geometric center of the particle. The “ghost” particle interacts only through dipolar forces, and its dipole is constrained to remain parallel to the rigid bond, pointing radially outward. In the limit $d = 0$, this model reduces to the standard Stockmayer fluid. This representation should be appropriate for most physically relevant ranges of shift and dipole magnitudes. Throughout this work, we vary the dipole magnitude $\mu = |\mu|$, shift d , and temperature T .

To avoid numerical discontinuities at the LJ cutoff, we apply a constant energy shift S to the potential,

$$\phi_{\text{LJ}}^{\text{shifted}}(r_{ij}) = \begin{cases} 4\epsilon \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^6 \right] + S & \text{if } r_{ij} \leq r_c, \\ 0 & \text{if } r_{ij} > r_c, \end{cases} \quad (3)$$

where S is chosen such that $\phi_{\text{LJ}}^{\text{shifted}}(r_c) = 0$. Dipolar interactions are evaluated directly in real space up to a cutoff distance of 8.0σ , beyond which they are computed in Fourier space using the particle-particle particle-mesh (PPPM) method for long-range dipolar interactions.⁴⁷

To determine vapor-liquid coexistence conditions, we employ slab-geometry simulations with box dimensions $10\sigma \times 10\sigma \times 100\sigma$ (such that $L_z = 10L_x = 10L_y$) containing 3000 physical particles (6000 interaction sites, including the LJ particles and their associated dipolar “ghosts”), as illustrated in Fig. 2. Initial configurations are generated by randomly placing particles in the simulation box using Packmol.⁴⁸ All simulations employ a Nosé-Hoover-type thermostat that accounts for both translational and rotational degrees of freedom in rigid bodies,^{49–52} with a relaxation time of $\tau = 100\Delta t$ and a time step of $\Delta t = 0.005\tau_0$, where $\tau_0 = \sqrt{m\sigma^2/\epsilon}$.

To induce slab formation, a weak external force is applied along the z -axis to all particles for 5×10^4 steps, driving them toward the center of the simulation cell. Similar to prior studies,^{26,53–55} to suppress interactions between the liquid slab and its periodic images, we

apply a slab correction by introducing 12-6 Lennard-Jones walls at the two ends of the simulation box shown in Fig. 2, together with a vacuum region three times the box length between periodic replicas (or $L_{\text{vac.}} = 3L_z$), imposing vacuum boundary conditions, thereby rendering the system periodic in two dimensions and nonperiodic in the normal dimension.⁵⁶ This slab correction adequately removes spurious interactions between liquid slabs along the z -direction while retaining the efficiency of the three-dimensional PPPM solver; more efficient correction methods are available in other simulation suites.^{57–61} We have repeated these simulations with varying vacuum size to confirm that the observables obtained, such as system energy and liquid densities, are invariant. After slab formation, the external force is removed, and the system is allowed to equilibrate for an additional 5×10^4 steps. Production simulations are then performed for 1×10^7 steps, with particle coordinates and dipole moment vectors recorded every 100 steps. We determine the bulk liquid density from the resulting phase-separated configurations at different values of μ and d . Based on the dipole autocorrelation functions and the liquid densities obtained, we concluded that this simulation procedure was sufficient to fully equilibrate the system.

To analyze the homogeneous liquid phase, we conduct bulk simulations at the coexistence liquid density determined from the slab calculations. The simulations are carried out in a cubic box with dimensions adjusted to maintain the coexistence liquid density for 3000 uniformly distributed particles. At large dipole moments, the liquid can reach very high densities, and simulations may diverge because of overlapping molecules, even following energy minimization. To mitigate this issue, we initialize the system with the box length in the x -direction set to 1.5 times the target value and gradually compress this dimension over 5×10^4 steps.⁶² This is followed by an additional 5×10^4 equilibration steps. Finally, production simulations are carried out for 1×10^7 steps, with particle coordinates and dipole moment vectors recorded every 100 steps. This simulation protocol is more than sufficient to accurately determine structural and dielectric properties.^{38,39,63}

2. Analysis

We analyze the simulation trajectories using the MDAnalysis toolkit⁶⁴ and MDCraft.⁶⁵ MDCraft is employed to accommodate the features of the shifted-dipole Stockmayer model by treating dipole orientations in addition to particle positions through a custom trajectory reader. In this work, we focus primarily on the interparticle positional and orientational correlations that determine the structural and dielectric properties of the system. These correlations are conveniently characterized using three key angles defined by Langenbach and co-workers,^{38,66} as illustrated in Fig. 3.

For any pair of particles 1 and 2, we define two angles ϕ_1 and ϕ_2 between the dipoles and the displacement vector connecting two real particle centers,

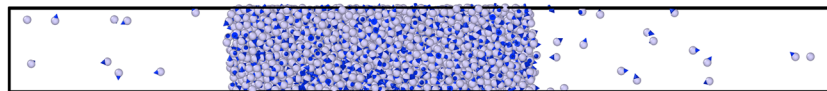


FIG. 2. Slab system for the simulation of the shifted-dipole Stockmayer particles. The Lennard-Jones beads are visualized in light purple, and the shifted dipoles are visualized in dark blue. Ghost particles have been hidden.

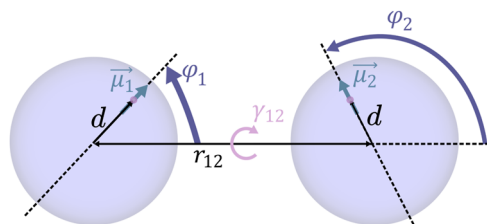


FIG. 3. Schematic illustration of angular descriptors for the specific particle pair. The angles φ_1 and φ_2 denote the orientation of each dipole moment $\vec{\mu}_1$ and $\vec{\mu}_2$ relative to the dipole-displacement vector $\vec{r}_{12}$, while γ_{12} is the torsional angle between the two dipoles. The scalar center-to-center distance is r_{12} , and d represents the shift of each dipole from the particle center.

$$\cos \varphi_1 \equiv \xi_1 = \frac{\vec{\mu}_1 \cdot \vec{r}_{12}}{|\vec{\mu}_1| |\vec{r}_{12}|}, \quad (4)$$

$$\cos \varphi_2 \equiv \xi_2 = \frac{\vec{\mu}_2 \cdot \vec{r}_{12}}{|\vec{\mu}_2| |\vec{r}_{12}|}, \quad (5)$$

together with the angle between the two dipole vectors,

$$\cos \Theta = \frac{\vec{\mu}_1 \cdot \vec{\mu}_2}{|\vec{\mu}_1| |\vec{\mu}_2|}. \quad (6)$$

The corresponding torsional angle γ_{12} is then defined as

$$\cos \gamma_{12} = \frac{\cos \Theta - \xi_1 \xi_2}{\sqrt{1 - (\xi_1)^2} \sqrt{1 - (\xi_2)^2}}. \quad (7)$$

From these quantities, we construct the multivariate pair distribution function $\rho(\xi_1, \xi_2, \gamma_{12}, r_{12})$, averaged over all particle pairs. Integration over selected degrees of freedom yields marginal distributions that allow specific structural features to be extracted. For example, the radial distribution function (RDF) is obtained as

$$g(r_{12}) = \frac{1}{4\pi r_{12}^2} \int d\xi_1 \int d\xi_2 \int d\gamma_{12} \rho(\xi_1, \xi_2, \gamma_{12}, r_{12}). \quad (8)$$

Oriental correlations resolved with respect to the intermolecular separation are similarly defined as

$$P(r_{12}, \xi_1) = \frac{1}{4\pi r_{12}^2} \int d\xi_2 \int d\gamma_{12} \rho(\xi_1, \xi_2, \gamma_{12}, r_{12}). \quad (9)$$

For Stockmayer-type fluids, it is also common to examine the angular distribution function (ADF),^{38,39}

$$g(\xi_1) = \int dr_{12} \int d\xi_2 \int d\gamma_{12} \rho(\xi_1, \xi_2, \gamma_{12}, r_{12}). \quad (10)$$

To isolate effects within the first solvation shell, the radial integrals are truncated at $r_{12} = 1.5\sigma$, which corresponds, on average, to the location of the first minimum in the RDF function. We note that this correlation function possesses a useful symmetry: while the ADF describes the angle between the dipole of particle 1 and the displacement vector between a pair, the corresponding distribution for particle 2 satisfies $g(\xi_2) = g(-\xi_1)$ [or $g(\varphi_2) = g(\pi - \varphi_1)$]. That is, it is simply the reflection of the first particle's ADF about $\xi = 0$ ($\varphi = \frac{\pi}{2}$). Without loss of generality, we, therefore, consider $g(\xi) \equiv g(\xi_1)$ [$g(\varphi) \equiv g(\varphi_1)$] hereafter. By a similar token, we also omit the indices on the center-of-mass separation distance r_{12} .

C. COFFEE theory

To gain deeper insights into the shifted-dipole Stockmayer fluid and to provide an analytical description of its equilibrium properties, we apply the COFFEE (Co-Oriented Fluid Functional Equation for Electrostatic interactions) theory developed by Langenbach and co-workers.^{38,39} A full derivation is provided in the original work; here, we summarize the equations relevant to our implementation.⁶⁷

The COFFEE framework expresses the residual Helmholtz free energy F_{res} of a system with Lennard-Jones and shifted dipolar interactions as the sum of three contributions,

$$\frac{F_{\text{res}}}{NkT} = \frac{F_{\text{FF}}}{NkT} + \frac{F_{\text{NF}}}{NkT} + \frac{F_{\text{ref}}}{NkT}, \quad (11)$$

where k is Boltzmann's constant and the subscripts FF, NF, and ref denote the far-field, near-field, and reference contributions, respectively.

In its original formulation, COFFEE theory employed the full Lennard-Jones fluid as the reference system.⁶⁸ Because our simulations use a truncated Lennard-Jones potential with cutoff $r_c = 2.5\sigma$, we instead adopt a reference free energy based on the perturbed truncated and shifted (PeTS) equation of state.⁶⁹

The dipolar free energy is separated into near- and far-field contributions under the assumption that shifting the dipole primarily affects interactions within a radius of 1.5σ .³⁸ The far-field term follows the functional form proposed by Vrabec and Gross,⁷⁰ employing a Padé approximation,⁷¹

$$F_{\text{FF}} = \frac{F_2}{1 - \frac{F_3}{F_2}}, \quad (12)$$

where F_2 and F_3 are parameterized from simulation data. The near-field contribution explicitly accounts for the local orientational environment of a reference particle through the orientational distribution function,

$$O(\xi_1, \xi_2, \gamma_{12}) = \frac{1}{Q} \exp \left(-\frac{24}{19} I_{\mu\mu}^{\text{NF}} \int dr_{12} \beta \phi_{\text{dipole}}(r_{12}, \xi_1, \xi_2, \gamma_{12}) \right). \quad (13)$$

In Eq. (13), the normalization constant Q , interpreted as an orientational partition function, is defined as

$$Q = \int d\xi_1 d\xi_2 d\gamma_{12} \times \exp \left(-\frac{24}{19} I_{\mu\mu}^{\text{NF}}(y^*) \int dr_{12} \beta \phi_{\text{dipole}}(r_{12}, \xi_1, \xi_2, \gamma_{12}) \right), \quad (14)$$

where the correction factor $I_{\mu\mu}^{\text{NF}}(y^*)$ depends only on the reduced dipole strength y^* ⁴¹ and is given by

$$I_{\mu\mu}^{\text{NF}}(y^*) = a_1 \exp(a_2 y^* + a_3 y^{*2}), \quad (15)$$

with

$$y^* = \frac{4\pi\beta\mu^2\rho}{9}. \quad (16)$$

Note that all radial integrals within the preceding equations are bound between $r = 0$ and $r = 1.5\sigma$, only considering the first solvation shell. Because a wider range of dipole magnitudes μ and shifts

d are explored here, and a different reference system is employed, the constants a_n are refitted using angular distribution functions obtained from simulation.

From Q , the near-field free-energy contribution can be written as⁷²

$$\frac{F_{\text{NF}}}{NkT} = -\frac{19\pi}{12}\rho\sigma^3 g_{\text{HS}}(\sigma) \ln \frac{Q}{\Omega}, \quad (17)$$

where $g_{\text{HS}}(\sigma)$ is the hard-sphere radial distribution function at contact.⁷³ Here, Ω denotes the total orientational phase-space volume; for uniformly distributed orientations on the unit sphere, $\Omega = 4\pi$.

The resulting theory, which we refer to as COFFEE-PeTS, is implemented in the open-source package Clapeyron.jl.⁶⁷ All fitted parameters are reported in the [supplementary material](#).

III. RESULTS AND DISCUSSION

We now examine the structural properties of the shifted-dipole Stockmayer fluid and the resulting dielectric behavior of homogeneous bulk phases. On the basis of these results, we subsequently analyze the vapor–liquid equilibrium and assess predictions obtained from the modified COFFEE equation.

Throughout this section, reduced Lennard-Jones units are employed. Accordingly, we define the dimensionless density, temperature, dipole moment, and shift as

$$\rho^* = \rho\sigma^3, \quad T^* = \frac{kT}{\varepsilon}, \quad \mu^* = \frac{\mu}{\sqrt{4\pi\varepsilon_0\varepsilon\sigma^3}}, \quad d^* = \frac{d}{\sigma}, \quad (18)$$

where the superscript $*$ denotes reduced quantities. As a reference, if $\varepsilon/k = 300$ K and $\sigma = 0.275$ nm, then $\mu^* \approx 1.95$ and $\rho^* = 0.9$ correspond approximately to liquid water near room temperature at $T^* = 1$.

A. Structural properties

A central objective of this study is to elucidate how shifting the dipole modifies the structure of the Stockmayer fluid. We, therefore, analyze several pair correlation functions: (1) the radial distribution function $g(r)$, (2) the angular–radial distribution $P(r, \varphi)$, (3) the angular dependence of dipole alignment $\langle \cos \Theta \rangle$ within the first solvation shell, and (4) the angular distribution function $g(\varphi)$. Note that all results obtained here pertain to the liquid state of the sd-SF under vapor–liquid equilibrium conditions.

The radial distribution function $g(r)$ describes the relative probability of finding a particle at a distance r from a reference particle. For the shifted-dipole Stockmayer fluid, $g(r)$ reflects the combined influence of excluded-volume repulsion and dipolar forces. By comparing $g(r)$ for different dipole magnitudes and shifts, we assess how breaking the particle symmetry modifies short-range order. For an unshifted-dipole Stockmayer fluid, the radial structure is expected to be dominated by repulsive interactions,¹⁷ such that $g(r)$ exhibits pronounced peaks near the integer values of r/σ . This behavior is confirmed in Fig. 4(a), where sharp oscillations indicate well-defined solvation shells. Increasing μ^* enhances local ordering, as reflected by the sharpening of the first peak and amplified subsequent oscillations. At intermediate dipole shifts, the radial distribution remains well defined, as shown in Fig. 4(b) for $d^* = 0.25$. As in the unshifted case, excluded-volume effects largely govern the

radial ordering. This is further emphasized, as shown in Fig. 4(c), which demonstrates that varying the dipole shift produces only minor changes in $g(r)$.

Although the dipole shift does not substantially alter radial ordering, it introduces pronounced angular anisotropy in the solvation shells owing to the directional nature of the dipolar field. For small μ^* and minimal shift, excluded-volume effects dominate and the shells remain nearly spherical, consistent with the isotropic pattern in Fig. 4(d). As μ^* increases, neighboring particles preferentially occupy regions near $\varphi = 0$ and $\varphi = \pi$, where dipolar interactions are strongest, leading to the angular modulation observed in Figs. 4(e) and 4(f). Note that this anisotropy, even at the largest dipole strength, only persists within the first solvation shell. Distributions for separations beyond 1.5σ are largely isotropic.

To relate these positional preferences to orientational correlations, Fig. 5 shows the variation of $\langle \cos \Theta \rangle$ with angular position within the first solvation shell, since this region exhibits the strongest anisotropy; this will also be the focus of all subsequent analyses of angular distributions. For unshifted dipoles [Fig. 5(a)], the distribution is symmetric about $\varphi = \pi/2$, reflecting the fore–aft symmetry of the dipolar field. Maxima in $\langle \cos \Theta \rangle$ appear near $\varphi = 0$ and $\varphi = \pi$, where the field aligns with the reference dipole and strengthens with increasing μ^* . Near $\varphi = \pi/2$, the field is weaker and favors anti-parallel alignment, producing a minimum in $\langle \cos \Theta \rangle$. At sufficiently large μ^* , this minimum becomes positive, indicating global ferroelectric ordering in the liquid.

Introducing a dipole shift breaks this symmetry, as illustrated in Figs. 4(g)–4(i). The field near $\varphi = 0$ becomes stronger than near $\varphi = \pi$, producing preferential accumulation at the head of the particle and broader angular distributions at the tail. Despite this increased ordering, as it was in the unshifted systems, these anisotropies only persist within the first solvation shell. Within the first solvation shell, these positional changes manifest in Fig. 5(b): despite higher local density near $\varphi = 0$, the average alignment is reduced, revealing orientational frustration induced by the shift. The effect is even more pronounced near $\varphi = \pi$.

At larger μ^* , an unexpected peak develops near $\varphi = 3\pi/4$ in Fig. 5(b). As schematically illustrated in Fig. 6, shifting the dipole displaces the region of the strongest electric field away from $\varphi = \pi$. Neighboring particles, therefore, accumulate slightly off axis, where the field is strongest, broadening the distribution around $\varphi = \pi$ and producing a secondary maximum, which becomes more pronounced at stronger dipole strengths. For the modest range of parameters considered in this work, this local ordering is largely isolated to the first solvation shell. As discussed by Kantorovich and co-workers,^{28,29} for larger dipole magnitudes and shifts, this ordering can extend beyond the first solvation shell, leading to the formation of larger clusters.

These trends are reflected in the angular distribution functions (ADF) $g(\varphi)$ shown in Fig. 7. Without a shift [Fig. 7(a)], the ADF is symmetric about $\varphi = \pi/2$, consistent with Figs. 4 and 5. Increasing μ^* enhances the probability of finding neighbors near $\varphi = 0$ and $\varphi = \pi$. Introducing a finite shift [Fig. 7(b)] breaks this symmetry, first amplifying $g(\varphi)$ near $\varphi = 0$ and subsequently producing a secondary maximum near $\varphi = 3\pi/4$. A weaker peak near $\varphi = \pi/4$ also emerges at large μ^* , consistent with the flattened angular distributions in Figs. 4(g)–4(i) and the reduced alignment seen in Fig. 5. Following

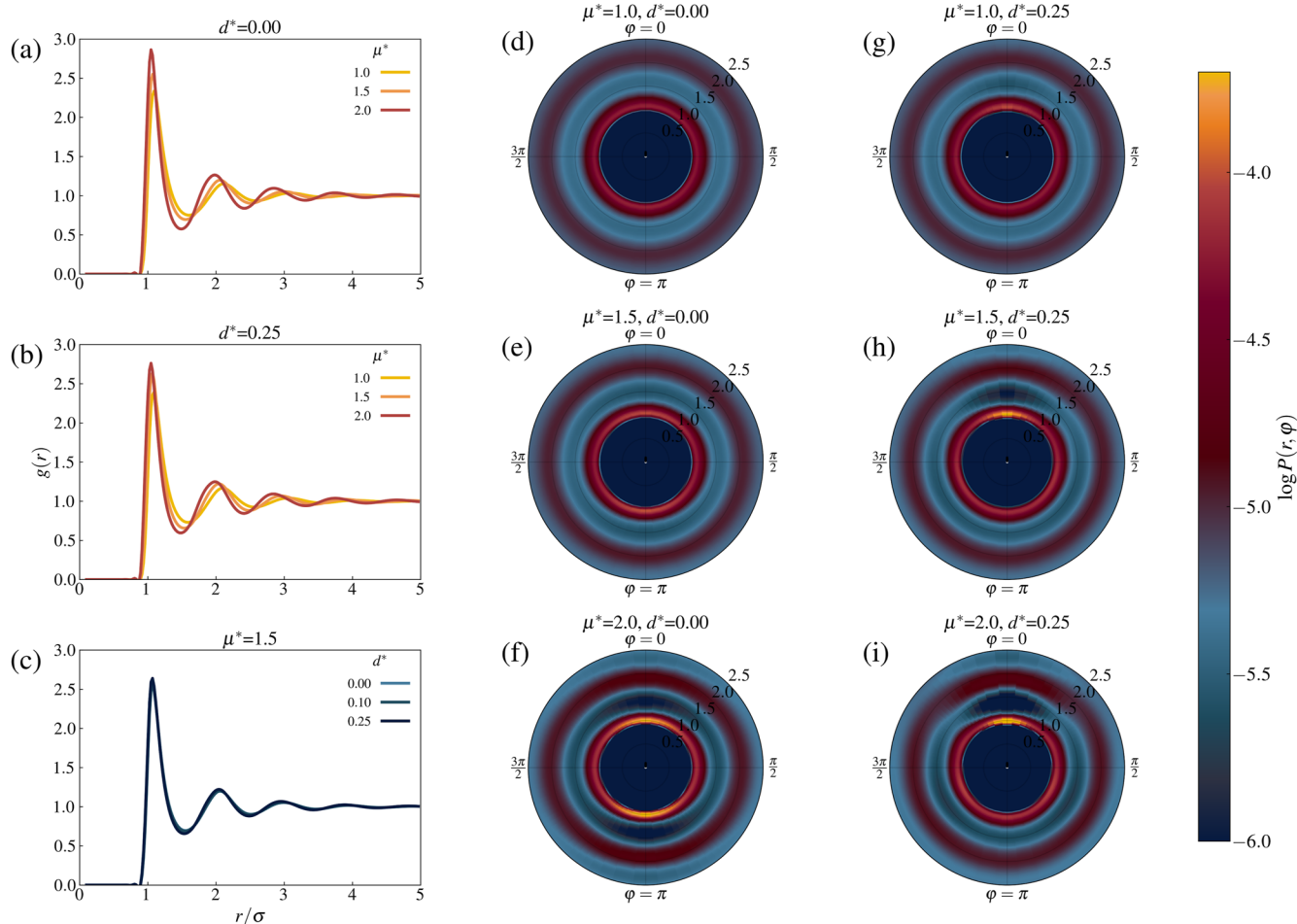


FIG. 4. Radial distribution functions $g(r)$ in panels (a)–(c) and angular–radial distribution functions $\log P(r, \varphi)$ in panels (d)–(i) for the shifted-dipole Stockmayer fluid. In (a) and (b), $d^* = 0.00$ and $d^* = 0.25$, respectively, with varying dipole strengths; panel (c) fixes $\mu^* = 1.500$ and varies d^* . Angular–radial distribution functions $\log P(r, \varphi)$ are in polar coordinates. Color denotes the log-normalized probability of finding a neighbor LJ center at separation r/σ and angle φ from a reference particle's dipole.

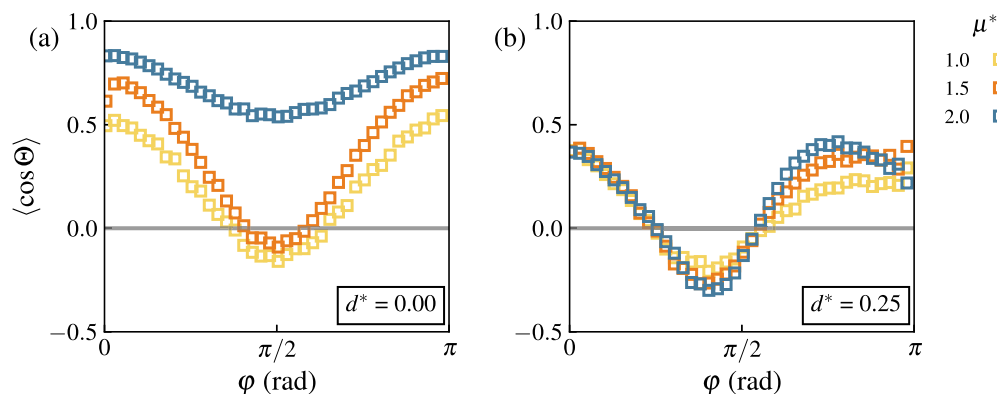


FIG. 5. Average angle between dipoles of neighboring particles, $\langle \cos \Theta \rangle$, in the first solvation shell as a function of angular position φ between a reference particle dipole and its neighboring particles for (a) $d^* = 0.001$ with $\mu^* = 1.000, 1.500, 2.000$ and (b) $d^* = 0.250$ with the same μ^* .

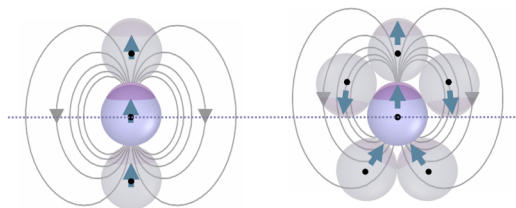


FIG. 6. Schematic of the dipole field around a tagged particle for a regular Stockmayer fluid particle (left) and a shifted-dipole Stockmayer fluid particle (right); the dotted midline is a visual guide to emphasize the shift. Black dots mark neighbor positions, with neighboring dipoles aligning with the local field.

from Fig. 5, we can see a minimum in $\langle \cos \Theta \rangle$ around these orientations, implying that these dipoles have a preference to be oriented anti-aligned with the reference particle. This is consistent with Fig. 6, as, in this region, the dipolar field is aligned such that the preferred orientation would be anti-aligned with the reference dipole.

Overall, introducing a dipole shift disrupts the preferential axial ordering characteristic of the conventional Stockmayer fluid at high dipole strengths, most clearly illustrated by the difference between Figs. 5(a) and 5(b) at $\mu^* = 2.0$, where $\langle \cos \Theta \rangle$ was previously > 0 for all φ , with a shift, values < 0 persist.

Finally, Fig. 8 illustrates the temperature dependence of the ADF at fixed $\mu^* = 1.5$. Because these data correspond to coexistence liquid densities, increasing temperature is accompanied by a decrease in density. The reduced packing facilitates accumulation near energetically favorable orientations at $\varphi = 0$ and $\varphi = \pi$, consistent with previous work.³⁸ For shifted-dipole Stockmayer fluids, the ADF flattens, likely reflecting diminished steric constraints at lower densities.

B. Dielectric properties

A key material property that depends strongly on dipolar correlations is the relative permittivity (dielectric constant), ϵ_r . The dielectric constant provides a macroscopic measure of the collective response of the fluid to an applied electric field and, therefore,

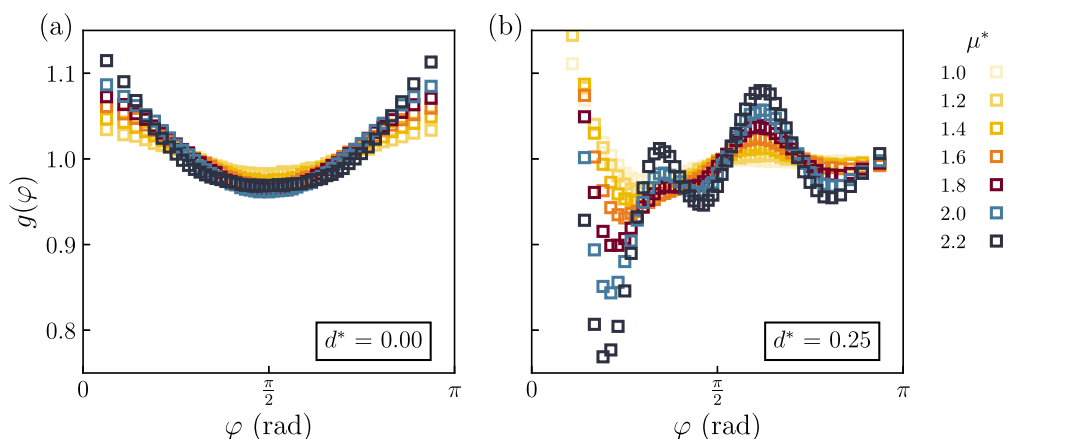


FIG. 7. Angular distribution function $g(\varphi)$ obtained from simulations for varying dipole strength at constant shift: (a) $d^* = 0.00$ and (b) $d^* = 0.25$. All systems were simulated at $T^* = 1.0$ and at the coexistence liquid density.

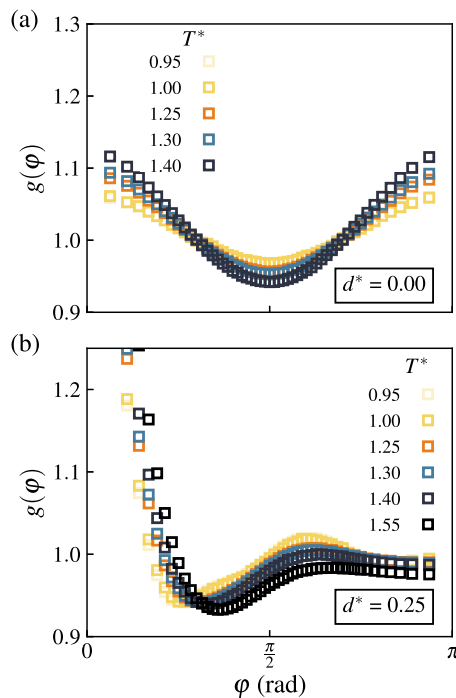


FIG. 8. Angular distribution function obtained from simulations for systems with varying temperature and dipole shift: (a) $d^* = 0.00$ and (b) $d^* = 0.25$. All systems were simulated with $\mu^* = 1.5$ and at the coexistence liquid density.

reflects the extent of orientational correlations in the dipolar liquid. In this section, we examine how ϵ_r depends on the dipole strength μ^* and shift d^* . We compute ϵ_r from equilibrium polarization fluctuations using the Kirkwood fluctuation formula,⁷⁴

$$\epsilon_r = 1 + \frac{4\pi\rho}{3kT} (\langle \mathbf{M}^2 \rangle - \langle \mathbf{M} \rangle^2), \quad (19)$$

where the total system dipole moment is

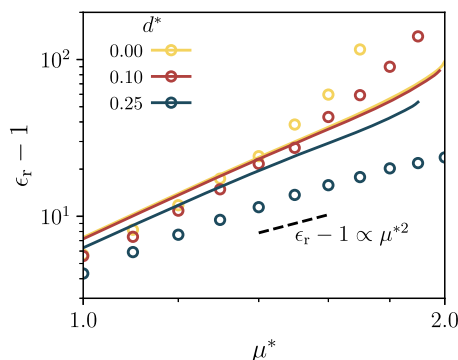


FIG. 9. Dielectric constant of the varying dipole strength (μ^*) and shift (d^*) at $T^* = 1$ and liquid density corresponding to vapor–liquid equilibrium conditions. Solid lines represent predictions made by COFFEE-PeTS, while symbols were obtained from molecular dynamics simulations. Error bars obtained from block-averaging of the dielectric constant were found to be negligible and are thus omitted.

$$\mathbf{M} = \sum_i \mu_i \quad (20)$$

For an isotropic, disordered fluid, one expects $\langle \mathbf{M} \rangle \approx 0$. Figure 9 reports ϵ_r obtained from the bulk simulations performed at $T^* = 1$ and the liquid densities at coexistence. For all cases, ϵ_r increases with the dipole strength μ^* , reflecting enhanced dipole–dipole correlations and stronger orientational ordering as μ^* increases. At the largest dipole moments, ϵ_r rises sharply. This could be attributed to a ferroelectric transition. However, the system size is too small to fully resolve this transition.

Because the dipole shifting increases the local interaction asymmetry, introducing a finite shift d^* disrupts the cooperative dipole alignment and leads to a systematic reduction in ϵ_r . In the limit of large shifts, ϵ_r approaches the mean-field result for noninteracting dipoles (the Debye–Langevin limit), shown as a dashed line in Fig. 9. This suggests that, for sufficiently large d^* , local dipole rearrangements do not build a strong self-consistent reaction field, and the dielectric response is dominated by the single-particle orientational susceptibility rather than collective correlations. This suppression is consistent with the structural trends discussed in Sec. III A: increasing d^* reduces the average alignment between neighboring

dipoles (Fig. 5) and broadens the angular distributions (Fig. 7), as schematically illustrated in Fig. 6.

Using the reparameterized COFFEE-PeTS model, we also predict ϵ_r at these state points. Although quantitative agreement deteriorates at large μ^* and d^* (conditions beyond the regime for which COFFEE was originally developed), the theory reproduces the qualitative trends and highlights the direct connection between the local orientational structure and the macroscopic dielectric response.

C. Vapor-liquid equilibria

Linking microscopic structure to macroscopic behavior, we examine the coexistence liquid densities obtained from simulation and compare them with predictions from COFFEE-PeTS in Fig. 10. We restrict the application of COFFEE-PeTS to intermediate values of the dipole strength and shift, since larger values lie beyond the regime for which the original COFFEE theory was developed.

Considering first the temperature dependence in Fig. 10(a), the coexistence liquid density increases systematically with dipole strength at a given temperature. Stronger dipoles enhance cohesive interactions, promoting closer packing and raising the critical temperature. These trends are consistent with previous studies⁷⁰ and are reproduced well by COFFEE-PeTS within its intended range of applicability. The introduction of a dipole shift, through competition between enhanced local interactions and the orientational frustration, results in a modest increase in the coexisting liquid density. This balance is also captured qualitatively by COFFEE-PeTS. At larger shifts, however, the quantitative performance of the theory deteriorates, as is particularly evident in Fig. 10(b). This deterioration will only worsen for larger shifts and dipole magnitude once longer-range structures form. For the purposes of this work, COFFEE-PeTS reproduces coexistence densities with high fidelity only for small d^* and moderate μ^* .

At very large dipole strengths and small displacements, Fig. 10(b) shows that increasing d^* initially leads to a decrease in the liquid density. At such high μ^* , the regular Stockmayer fluid develops strong global polarization and highly ordered local packing, which results in unusually dense liquid phases, consistent with Fig. 5 and previous studies. Introducing even a small displacement

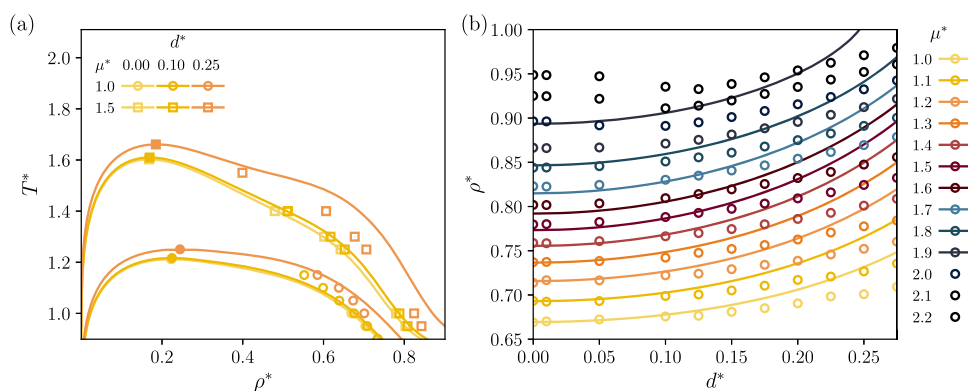


FIG. 10. Liquid density at coexistence with varying dipole strength, shift, and temperature. Panel (a) examines the full vapor–liquid envelope with increasing temperature, while panel (b) considers the liquid density branch only at $T^* = 1.0$ and varying μ^* and d^* . Solid lines represent predictions made by COFFEE-PeTS, while symbols were obtained from molecular dynamics simulations.

disrupts this ordering, suppressing the formation of the polarized state and thereby reducing the liquid density.

Considering Figs. 9 and 10 together reveals an intriguing result. Although introducing a dipole shift effectively strengthens cohesive interactions and increases the liquid density, the dipolar correlations, and consequently the dielectric constant, are suppressed. This behavior is counterintuitive: according to the Kirkwood fluctuation formula, stronger intermolecular interactions and denser packing would typically enhance collective dipole fluctuations and thus increase the dielectric constant. Using the physical intuition developed in Sec. III A, we can rationalize this behavior: while the dipole shift increases the magnitude of local interactions, it simultaneously introduces orientational frustration within the first solvation shell, thereby weakening cooperative dipole alignment and suppressing long-range polarization.

IV. CONCLUSION

In this work, we have combined molecular simulations with analytical theory to demonstrate that even a modest shift of the dipole off the center of a Stockmayer particle can profoundly alter its microscopic structure, dielectric response, and macroscopic phase behavior. Although the aspects of shifted dipoles have been explored previously, our results provide a unified physical picture that connects local symmetry breaking in pair correlations to emergent thermodynamic consequences.

The central effect of shifting the dipole is the loss of fore-aft symmetry in the local environment surrounding each particle. While radial distribution functions remain largely governed by excluded-volume packing, the angular structure of the solvation shell changes dramatically: the dipolar field is strengthened at the head of the particle ($\varphi = 0$) and weakened at the tail ($\varphi = \pi$), producing preferential accumulation and alignment in the forward direction and frustrated orientational correlations in the rear. At sufficiently strong dipoles, this frustration becomes particularly pronounced, shifting preferred configurations away from axial positions and disrupting the highly ordered structural characteristic of the conventional Stockmayer fluid.

These microscopic asymmetries propagate directly to collective electrostatic behavior. Despite stronger local interactions, introducing a dipole shift weakens cooperative dipolar correlations, leading to a systematic reduction in the dielectric constant. For large shifts, the dielectric response approaches the mean-field Debye limit, indicating that local dipole rearrangements no longer generate a strong self-consistent reaction field and that collective polarization is effectively suppressed, reducing the dielectric constant by an order of magnitude.

The consequences of dipolar asymmetry extend to phase behavior. Increasing dipole strength enhances cohesion and raises the critical temperature, as in the standard Stockmayer model, but the dipole shift fundamentally alters this trend. At intermediate dipole strengths, the introduction of a shift further enhances cohesion and raises the liquid density. However, at high dipole strengths, where the unshifted fluid develops strongly polarized, densely packed liquid phases, the introduction of even a small shift frustrates this ordering, suppresses the formation of a ferroelectric-like state, and lowers the liquid density. Conversely, for disordered fluid phases, the increased attraction between shifted dipoles increases the liquid

density. In this way, the dipole shift acts not simply as a quantitative perturbation but as a qualitative control parameter governing structure, dielectric response, and coexistence.

Taken together, our results reveal a striking principle: modifying the *geometry* of electrostatic interactions can be more disruptive to the liquid structure than the raw magnitude of the interactions themselves. This sensitivity to dipole placement opens new avenues for designing model fluids and coarse-grained representations of polar materials in which dielectric properties, phase behavior, and interfacial structure can be tuned independently. Such control is likely to be relevant for a broad class of systems, including polar solvents, electrolytes, ionic liquids, and soft interfaces, where displaced charges and anisotropic electrostatic environments are ubiquitous. Future work will extend these ideas to multicomponent mixtures and confined geometries, where the geometric frustration of electrostatics may play an even more dramatic role.

SUPPLEMENTARY MATERIAL

The [supplementary material](#) contains details regarding our simplification of the near-field term for the COFFEE equation, as well as the refitted parameters for our COFFEE-PeTS implementation.

ACKNOWLEDGMENTS

The authors would like to thank Dr. Benjamin Ye for helping develop much of the tools required for the analysis used in this work. S.V. is supported by the U.S. Department of Energy, Office of Science, Office of Advanced Scientific Computing Research, Department of Energy Computational Science Graduate Fellowship, under Award No. DE-SC0022158. Partial support for this research is provided by Hong Kong Quantum AI Lab, AIR@InnoHK of Hong Kong Government. B.Z. gratefully acknowledges support by Harvey Mudd College start-up grant and the NSF under Grant Nos. CHE-2337602 and OAC-2408259. A.V. acknowledges the Resnick Sustainability Institute and the WAVE Fellows Program at Caltech for their support.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Pierre J. Walker, Ananya Venkatachalam, and Samuel Varner contributed equally to this paper.

Pierre J. Walker: Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Software (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (equal). **Ananya Venkatachalam:** Data curation (equal); Formal

analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Samuel Varner**: Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Software (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (equal). **Bilin Zhuang**: Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Project administration (equal); Supervision (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Zhen-Gang Wang**: Conceptualization (equal); Funding acquisition (equal); Investigation (equal); Project administration (equal); Resources (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

- 1 B. Zhuang, G. Ramanauskaitė, Z. Y. Koa, and Z.-G. Wang, *Sci. Adv.* **7**, eabe7275 (2021).
- 2 I. Nakamura, A.-C. Shi, and Z.-G. Wang, *Phys. Rev. Lett.* **109**, 257802 (2012).
- 3 K. P. Gregory, G. R. Elliott, H. Robertson, A. Kumar, E. J. Wanless, G. B. Webber, V. S. J. Craig, G. G. Andersson, and A. J. Page, *Phys. Chem. Chem. Phys.* **24**, 12682 (2022).
- 4 A. M. Jubb, W. Hua, and H. C. Allen, *Annu. Rev. Phys. Chem.* **63**, 107 (2012).
- 5 D. J. Tobias, A. C. Stern, M. D. Baer, Y. Levin, and C. J. Mundy, *Annu. Rev. Phys. Chem.* **64**, 339 (2013).
- 6 M. F. Ruiz-Lopez, J. S. Francisco, M. T. C. Martins-Costa, and J. M. Anglada, *Nat. Rev. Chem.* **4**, 459 (2020).
- 7 F. Tang, T. Ohto, S. Sun, J. R. Rouxel, S. Imoto, E. H. G. Backus, S. Mukamel, M. Bonn, and Y. Nagata, *Chem. Rev.* **120**, 3633 (2020).
- 8 M. T. C. Martins-Costa and M. F. Ruiz-López, *J. Am. Chem. Soc.* **145**, 1400 (2023).
- 9 M. T. C. Martins-Costa and M. F. Ruiz-López, *Angew. Chem., Int. Ed.* **64**, e202418593 (2025).
- 10 Q. Liang, C. Zhu, and J. Yang, *J. Am. Chem. Soc.* **145**, 10159 (2023).
- 11 R. Xie, K. Guo, Y. Li, Y. Zhang, H. Zhong, D. Y. C. Leung, and H. Huang, *Nat. Commun.* **15**, 8860 (2024).
- 12 J. Kang, S. Wang, C. Li, G. Cao, X. Gong, and C. Zhu, *Wiley Interdiscip. Rev.: Comput. Mol. Sci.* **15**, e70031 (2025).
- 13 L. Shi, R. A. LaCour, N. Qian, J. P. Heindel, X. Lang, R. Zhao, T. Head-Gordon, and W. Min, *Nature* **640**, 87 (2025).
- 14 A. M. Kol, T. Herriman, W. Salmon, G. DiLabio, and M. Jackson, “Long-range dipole alignment of molecules generates strong local electric fields at air–water interfaces,” *chemRxiv:10.26434/chemrxiv-2025-dh30c-v2* (2025).
- 15 J. C. Shirley, Z. X. Ng, K.-Y. Chiang, Y. Nagata, Y. Litman, A. S. Hazrah, and M. Bonn, *J. Am. Chem. Soc.* **147**, 46163–46173 (2025).
- 16 W. H. Stockmayer, *J. Chem. Phys.* **9**, 863 (1941).
- 17 C. J. Shock, M. J. Stevens, A. L. Frischknecht, and I. Nakamura, *J. Phys. Chem. B* **124**, 4598 (2020).
- 18 X. Liu and X. Kong, *J. Phys. Chem. B* **128**, 3953 (2024).
- 19 E. L. Pollock and B. J. Alder, *Physica A* **102**, 1 (1980).
- 20 D. J. Adams and E. M. Adams, *Mol. Phys.* **42**, 907 (1981).
- 21 B. Groh and S. Dietrich, *Phys. Rev. E* **50**, 3814 (1994).
- 22 J. Bartke and R. Hentschke, *Mol. Phys.* **104**, 3057 (2006).
- 23 J. Eggebrecht, S. M. Thompson, and K. E. Gubbins, *J. Chem. Phys.* **86**, 2299 (1987).
- 24 S. Paul and A. Chandra, *J. Phys. Chem. B* **107**, 12705 (2003).
- 25 S. Samin, Y. Tsori, and C. Holm, *Phys. Rev. E* **87**, 052128 (2013).
- 26 S. G. Moore, M. J. Stevens, and G. S. Grest, *Phys. Rev. E* **91**, 022309 (2015).
- 27 I. Kusaka, Z.-G. Wang, and J. H. Seinfeld, *J. Chem. Phys.* **102**, 913 (1995).
- 28 S. Kantorovich, R. Weeber, J. J. Cerdà, and C. Holm, *Soft Matter* **7**, 5217 (2011).
- 29 S. Kantorovich, R. Weeber, J. J. Cerdà, and C. Holm, *J. Magn. Magn. Mater.* **323**, 1269 (2011), a part of Special Issue: Proceedings of 12th International Conference on Magnetic Fluid.
- 30 M. Klinkigt, R. Weeber, S. Kantorovich, and C. Holm, *Soft Matter* **9**, 3535 (2013).
- 31 R. Weeber, M. Klinkigt, S. Kantorovich, and C. Holm, *J. Chem. Phys.* **139**, 214901 (2013).
- 32 L. Baraban, D. Makarov, O. G. Schmidt, G. Cuniberti, P. Leiderer, and A. Erbe, *Nanoscale* **5**, 1332 (2013).
- 33 G. Steinbach, D. Nissen, M. Albrecht, E. V. Novak, P. A. Sánchez, S. S. Kantorovich, S. Gemming, and A. Erbe, *Soft Matter* **12**, 2737 (2016).
- 34 L. Rossi, J. G. Donaldson, J.-M. Meijer, A. V. Petukhov, D. Kleckner, S. S. Kantorovich, W. T. M. Irvine, A. P. Philipse, and S. Sacanna, *Soft Matter* **14**, 1080 (2018).
- 35 G. Steinbach, M. Schreiber, D. Nissen, M. Albrecht, E. Novak, P. A. Sánchez, S. S. Kantorovich, S. Gemming, and A. Erbe, *Phys. Rev. E* **100**, 012608 (2019).
- 36 G. Steinbach, M. Schreiber, D. Nissen, M. Albrecht, S. Gemming, and A. Erbe, *Phys. Rev. Res.* **2**, 023092 (2020).
- 37 K. Langenbach, C. Engin, S. Reiser, M. Horsch, and H. Hasse, *AIChE J.* **61**, 2926 (2015).
- 38 K. Langenbach, *Chem. Eng. Sci.* **174**, 40 (2017).
- 39 M. Kohns, J. Marx, and K. Langenbach, *J. Chem. Eng. Data* **65**, 5891 (2020).
- 40 M. Kohns, J. Marx, and K. Langenbach, *Chem. Eng. Sci.* **245**, 116875 (2021).
- 41 J. Marx, M. Kohns, and K. Langenbach, *Chem. Ing. Tech.* **94**, 1345 (2022).
- 42 J. Marx, M. Kohns, and K. Langenbach, *Fluid Phase Equilib.* **568**, 113742 (2023).
- 43 S. Varner, P. J. Walker, A. Venkatachalam, B. Zhuang, and Z.-G. Wang, *J. Chem. Phys.* **164**, 184511 (2026).
- 44 A. P. Thompson, H. M. Aktulga, R. Berger, D. S. Bolintineanu, W. M. Brown, P. S. Crozier, P. J. in ’t Veld, A. Kohlmeyer, S. G. Moore, T. D. Nguyen, R. Shan, M. J. Stevens, J. Tranchida, C. Trott, and S. J. Plimpton, *Comput. Phys. Commun.* **271**, 108171 (2022).
- 45 W. M. Brown, P. Wang, S. J. Plimpton, and A. N. Tharrington, *Comput. Phys. Commun.* **182**, 898 (2011).
- 46 W. M. Brown, A. Kohlmeyer, S. J. Plimpton, and A. N. Tharrington, *Comput. Phys. Commun.* **183**, 449 (2012).
- 47 J. J. Cerdà, V. Ballenegger, O. Lenz, and C. Holm, *J. Chem. Phys.* **129**, 234104 (2008).
- 48 L. Martínez, R. Andrade, E. G. Birgin, and J. M. Martínez, *J. Comput. Chem.* **30**, 2157 (2009).
- 49 H. Kamberaj, R. J. Low, and M. P. Neal, *J. Chem. Phys.* **122**, 224114 (2005).
- 50 W. G. Hoover, *Phys. Rev. A* **31**, 1695 (1985).
- 51 G. J. Martyna, M. E. Tuckerman, D. J. Tobias, and M. L. Klein, *Mol. Phys.* **87**, 1117 (1996).
- 52 G. J. Martyna, M. L. Klein, and M. Tuckerman, *J. Chem. Phys.* **97**, 2635 (1992).
- 53 V. A. Frolov and S. H. L. Klapp, *J. Chem. Phys.* **124**, 134701 (2006).
- 54 J. Richardi, M. P. Pileni, and J. J. Weis, *J. Chem. Phys.* **130**, 124515 (2009).
- 55 J. Jordanovic and S. H. L. Klapp, *Phys. Rev. E* **79**, 021405 (2009).
- 56 I.-C. Yeh and M. L. Berkowitz, *J. Chem. Phys.* **111**, 3155 (1999).
- 57 A. Arnold, J. de Joannis, and C. Holm, *J. Chem. Phys.* **117**, 2496 (2002).
- 58 A. Arnold and C. Holm, *Comput. Phys. Commun.* **148**, 327 (2002).
- 59 J. de Joannis, A. Arnold, and C. Holm, *J. Chem. Phys.* **117**, 2503 (2002).
- 60 G. Sutmann, P. Gibbon, and T. Lippert, *Fast Methods for Long-range Interactions in Complex Systems* (Forschungszentrum Jülich, 2011) google-Books-ID: GBnCcY22tBwC.

- ⁶¹A. Arnold, F. Fahrenberger, C. Holm, O. Lenz, M. Bolten, H. Dachsels, R. Halver, I. Kabadshow, F. Gähler, F. Heber, J. Iseringhausen, M. Hofmann, M. Pippig, D. Potts, and G. Sutmann, *Phys. Rev. E* **88**, 063308 (2013).
- ⁶²After further consultation of the Packmol documentation, it was found that the issues experienced in this work can be largely circumvented by specifying atom diameters in Packmol such that particle overlaps are avoided. Using this approach allows one to directly initialize the system without the divergences observed in this work.
- ⁶³C. J. Shock, M. J. Stevens, A. L. Frischknecht, and I. Nakamura, *J. Chem. Phys.* **159**, 134507 (2023).
- ⁶⁴N. Michaud-Agrawal, E. J. Denning, T. B. Woolf, and O. Beckstein, *J. Comput. Chem.* **32**, 2319 (2011).
- ⁶⁵B. B. Ye, P. J. Walker, and Z.-G. Wang, *J. Open Source Softw.* **9**, 7013 (2024).
- ⁶⁶K. Langenbach and M. Kohns, *J. Chem. Eng. Data* **65**, 980 (2020).
- ⁶⁷P. J. Walker, H.-W. Yew, and A. Riedemann, *Ind. Eng. Chem. Res.* **61**, 7130 (2022).
- ⁶⁸T. Lafitte, A. Apostolakou, C. Avendaño, A. Galindo, C. S. Adjiman, E. A. Müller, and G. Jackson, *J. Chem. Phys.* **139**, 154504 (2013).
- ⁶⁹M. Heier, S. Stephan, L. Jinlu, G. C. Walter, H. Hans, and K. Langenbach, *Mol. Phys.* **116**, 2083 (2018).
- ⁷⁰J. Gross and J. Vrabec, *AIChE J.* **52**, 1194 (2006).
- ⁷¹G. Stell, J. C. Rasaiah, and H. Narang, *Mol. Phys.* **27**, 1393 (1974).
- ⁷²Although this expression appears simpler than those originally presented by Langenbach and co-workers, it is analytically equivalent under the same assumptions. A full derivation is provided in the [supplementary material](#).
- ⁷³T. Boublik, *J. Chem. Phys.* **53**, 471 (1970).
- ⁷⁴M. Neumann, *Mol. Phys.* **50**, 841 (1983).